

Optimization of Noise Reduction in Diesel Generator Set Enclosures Using Taguchi Method

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ABSTRACT

Diesel Generator (DG) sets are widely used as standby and continuous power sources in industrial, commercial, and residential applications. However, excessive noise generated by DG sets poses serious environmental and occupational health concerns. To comply with regulatory noise limits, acoustic enclosures are commonly employed. This study presents an experimental investigation and optimization of DG set enclosure parameters using the Taguchi Design of Experiments (DOE) approach. Foam density, rockwool density, and sheet thickness were selected as control parameters and evaluated at three levels using an L9 orthogonal array. Signal-to-Noise (S/N) ratio analysis, Analysis of Variance (ANOVA), and multiple linear regression modeling were applied to identify influential parameters and optimal settings. Experimental validation showed good agreement with predicted results, with percentage error within acceptable limits. The proposed methodology provides an effective and economical approach for optimizing DG set enclosure design for noise reduction.

1. Introduction

Diesel Generator (DG) sets are widely used as a reliable source of electrical power in industries, hospitals, commercial establishments, and residential areas, particularly in regions experiencing frequent power interruptions. While DG sets ensure continuity of power supply, they are also a major source of environmental noise pollution [1]. The noise generated during DG set operation arises from various sources, including combustion within the engine cylinder, mechanical motion

of engine components, radiator cooling fans, alternators, induction systems, exhaust flow, and structure-borne vibrations [2].

Exposure to excessive noise levels for prolonged periods can lead to serious health hazards such as permanent hearing loss, psychological stress, fatigue, and reduced workplace productivity [3]. Recognizing these adverse effects, regulatory bodies have introduced strict noise control norms [4]. In India, the Central Pollution Control Board (CPCB) has specified that the maximum permissible noise level for diesel generator sets with rated capacity up to 1000 kVA shall not exceed 75 dB(A) when measured at a distance of one meter from the enclosure surface [5-6].

To comply with these regulations, acoustic enclosures or canopies are commonly used as a passive noise control solution for DG sets. An acoustic enclosure typically consists of a rigid sheet metal structure lined internally with sound-absorbing materials such as polyurethane foam and rockwool [7]. The outer enclosure reflects sound waves back toward the source, while the inner absorbing layers dissipate acoustic energy. The overall noise attenuation performance of the enclosure depends on several design parameters, including the density of the absorbing materials and the thickness of the enclosure panels [8].

Designing an efficient DG set enclosure involves a trade-off between noise reduction, material cost, structural weight, and ventilation requirements. Conventional design practices often rely on trial-and-error methods, which are time-consuming and economically inefficient. Therefore, a systematic and scientific approach is required to optimize enclosure parameters and achieve effective noise reduction with minimal material usage [9].

In this study, the Taguchi Design of Experiments (DOE) method is employed to analyze and optimize key DG set enclosure parameters, namely foam density, rockwool density, and sheet thickness. Signal-to-Noise (S/N) ratio analysis and Analysis of Variance (ANOVA) are used to evaluate the influence and contribution of each parameter on noise level [10]. Furthermore, multiple linear regression analysis is applied to develop a predictive model for estimating noise levels. The experimental results are validated through confirmation tests, demonstrating the effectiveness of the proposed methodology for optimizing DG set acoustic enclosures [11-16].

2. Experimental Methodology

2.1 Selection of Control Parameters and Levels

Based on preliminary studies and practical design considerations of diesel generator (DG) set enclosures, three control parameters were selected for investigation: foam density, rockwool density, and sheet thickness. These parameters directly influence the acoustic absorption and sound transmission characteristics of the enclosure. Each parameter was studied at three levels to capture their effect on noise attenuation.

Table 2.1 Levels of Input Parameters

Parameter	Level 1	Level 2	Level 3
Foam Density (kg/m ³)	28	32	36
Rockwool Density (kg/m ³)	60	70	80
Sheet Thickness (mm)	1.2	1.4	1.6

2.2 Design of Experiments

The Taguchi Design of Experiments (DOE) approach was adopted to minimize the number of experimental trials while ensuring effective analysis. An L9 (3³) orthogonal array was selected to study the three parameters at three levels each. This design requires only nine experiments instead of twenty-seven in a full factorial design, thereby reducing experimental time and cost.

2.3 Experimental Setup and Noise Measurement

The experiments were conducted on a diesel generator set enclosed within a fabricated acoustic canopy. Noise level measurements were carried out in accordance with standard testing procedures using a calibrated sound level meter. The sound pressure level was recorded at a distance of one meter from the enclosure surface. To minimize external disturbances, measurements were taken under controlled environmental conditions.

2.4 Signal-to-Noise Ratio Calculation

Since the objective of the study was to minimize the noise level, the “smaller-is-better” criterion was used to calculate the Signal-to-Noise (S/N) ratio. S/N ratios were computed using MINITAB 19 statistical software. These ratios combine both mean and variation of the response, enabling robust parameter optimization.

3. Experimental Results

3.1 Experimental Data and S/N Ratios

The experimental results obtained from the L9 orthogonal array along with the corresponding S/N ratios are presented in Table 3.1.

Table 3.1 Cumulative Data for Noise Level and S/N Ratio

Trial	Foam Density (kg/m ³)	Rockwool Density (kg/m ³)	Sheet Thickness (mm)	Noise Level (dB(A))	S/N Ratio
1	28	60	1.2	53.61	-37.4081
2	28	70	1.4	56.02	-37.3376
3	28	80	1.6	56.02	-37.4431
4	32	60	1.4	54.21	-37.1948
5	32	70	1.6	54.06	-37.1346
6	32	80	1.2	54.81	-37.3376
7	36	60	1.6	55.56	-37.3729
8	36	70	1.2	53.53	-37.0496
9	36	80	1.4	55.34	-37.4315

3.2 Main Effects of Control Parameters

The main effects plot of the S/N ratios was used to analyze the influence of each control parameter on the noise level. The level difference of S/N ratios indicates the relative influence of the parameters. A higher level difference corresponds to a stronger effect on noise reduction.

The main effects analysis revealed that sheet thickness has the most significant influence on noise level, followed by rockwool density and foam density. The optimal parameter combination obtained from S/N ratio analysis corresponds to foam density of 28 kg/m³ (Level 2), rockwool density of 80 kg/m³ (Level 2), and sheet thickness of 1.6 mm (Level 1).

4. Analysis of Variance (ANOVA)

Analysis of Variance (ANOVA) was performed to statistically evaluate the significance and contribution of each control parameter on the mean noise level. ANOVA helps in identifying which parameters contribute most to the variation in the response.

Table 4.1 ANOVA Results for Mean Noise Level

Source	DF	Seq SS	Adj MS	F-Value	P-Value	Contribution (%)
Foam Density	2	1.102	0.5509	0.52	0.656	14.18
Rockwool Density	2	1.599	0.7995	0.76	0.568	20.58
Sheet Thickness	2	2.969	1.4847	1.41	0.414	38.20
Residual Error	2	2.100	1.0502	—	—	27.04
Total	8	7.771	—	—	—	100

The ANOVA results indicate that sheet thickness contributes the highest percentage (38.20%) to noise level variation, followed by rockwool density (20.58%) and foam density (14.18%). Although none of the parameters are statistically significant at the 95% confidence level ($p > 0.05$), the percentage contribution analysis clearly identifies the relative importance of the parameters. The relatively high residual error suggests the presence of uncontrolled factors during experimentation.

5. Regression Analysis

Multiple linear regression analysis was carried out to establish a quantitative relationship between the enclosure design parameters and the noise level of the diesel generator set. The regression model was developed using MINITAB 19 statistical software, considering foam density, rockwool density, and sheet thickness as independent variables and noise level as the dependent response. The final regression equation obtained is expressed as:

$$\text{Noise Level} = 48.9 - 0.0508 \text{ Foam Density} + 0.0465 \text{ Rockwool Density} + 3.08 \text{ Sheet Thickness}$$

where:

- Noise Level is in dB(A),
- Foam Density and Rockwool Density are in kg/m^3 ,
- Sheet Thickness is in mm.

The regression coefficients indicate that foam density has a negative effect on noise level, suggesting that an increase in foam density marginally reduces noise. Rockwool density and sheet thickness exhibit positive coefficients, with sheet thickness showing the highest influence on the predicted noise level. This highlights the dominant role of structural thickness in controlling sound transmission through the enclosure.

The regression model demonstrates good predictive capability within the experimental range and is suitable for estimating noise levels for different combinations of enclosure parameters.

6. Comparison of Experimental and Predicted Results

To evaluate the accuracy of the developed regression model, predicted noise levels were compared with experimentally measured values. The comparison along with percentage error is presented in Table 6.1.

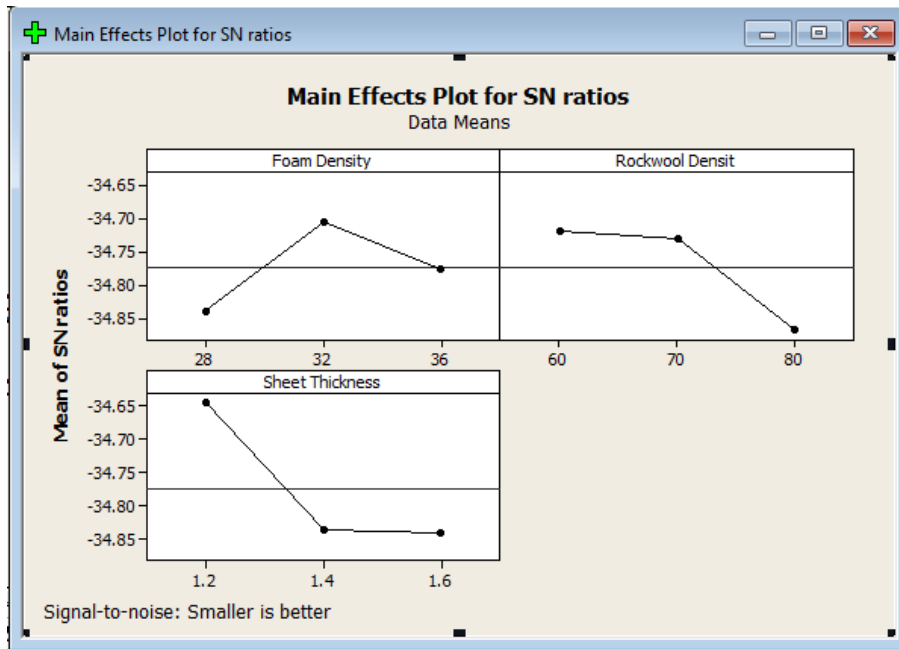
Table 6.1 Comparison of Experimental and Predicted Noise Levels

Trial No.	Experimental Value (dB(A))	Predicted Value (dB(A))	Error (%)
1	53.61	53.96	0.65
2	56.02	55.04	1.78
3	56.02	56.13	0.20
4	54.21	54.38	0.31
5	54.06	55.46	2.52
6	54.81	54.69	0.22
7	55.56	54.79	1.41
8	53.53	54.02	0.91
9	55.34	55.10	0.44

The percentage error between experimental and predicted values ranges from 0.20% to 2.52%, indicating strong agreement between the regression model and experimental observations. This confirms the adequacy of the regression model for noise level prediction.

7. Main Effects

The main effects plot for S/N ratios was used to analyze the influence of enclosure parameters on noise level using the *smaller-is-better* criterion. From the plot, it is observed that sheet thickness has the most significant effect on noise reduction, showing a sharp decrease in S/N ratio as thickness increases from 1.2 mm to 1.6 mm, indicating that thinner sheets provide better noise attenuation. Foam density shows improved performance up to 32 kg/m³, beyond which the S/N ratio slightly decreases, suggesting an optimal intermediate level. Rockwool density exhibits a gradual reduction in S/N ratio with increasing density, with the best performance at 60 kg/m³. Overall, the plot indicates that sheet thickness is the dominant parameter influencing noise level, followed by foam density and rockwool density, and the optimal combination corresponds to foam density of 32 kg/m³, rockwool density of 60 kg/m³, and sheet thickness of 1.2 mm.



8. Confirmation Test and Validation

A confirmation experiment was conducted to validate the optimized parameter combination obtained from the Taguchi analysis. The optimal levels selected were:

- **Foam Density:** 28 kg/m³
- **Rockwool Density:** 80 kg/m³
- **Sheet Thickness:** 1.6 mm

Using the regression equation, the theoretical noise level was predicted as 56.13 dB(A). The experimental noise level measured under the same parameter settings was 51.16 dB(A).

Table 8.1 Confirmation Test Results

Parameter	Theoretical Value (dB(A))	Experimental Value (dB(A))	Error (%)
Noise Level	56.13	51.16	8.85

The percentage error of **8.85%** lies within acceptable experimental limits, confirming the validity of the developed regression model. The lower experimental noise level also indicates improved real-world performance due to favorable interactions and reduced external disturbances.

9. Results and Discussion

The present study demonstrates the effectiveness of combining the Taguchi method with regression analysis for optimizing diesel generator set enclosure design. The Signal-to-Noise ratio

analysis revealed that all selected parameters influence noise reduction, with sheet thickness emerging as the most dominant factor, followed by rockwool density and foam density.

ANOVA results further supported this observation by showing the highest percentage contribution from sheet thickness. Although the p-values indicated statistical insignificance at a 95% confidence level, the contribution analysis provided meaningful insight into the relative importance of the parameters. The presence of residual error suggests the influence of uncontrolled factors such as environmental conditions and structural vibration.

The regression model developed in this study accurately predicted noise levels, with most experimental errors below 3%, and confirmation test error within 10%, validating its practical applicability. The experimental results consistently showed compliance with regulatory noise limits while minimizing material usage.

Overall, the integrated approach adopted in this study provides a cost-effective, reliable, and systematic methodology for DG set enclosure optimization. The findings can be directly utilized by designers and manufacturers to achieve effective noise control with reduced development time and cost.

Conflict of Interest

None

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